QTPTC1 Module 1

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Quantum Coins and Superposition

The purpose of this lab was to contrast classical randomness with quantum superposition, with reference to the IBM Quantum learning module on quantum coins. The classical coin flip was established with a simulated coin whose outcomes were distributed near $50-50$, which showed a process with no phase information and no coherence. In comparison, a single qubit prepared with a Hadamard gate was treated as a quantum coin. Circuits were transpiled for an IBM backend and were executed with the Sampler to obtain outcome counts and with the Estimator to access expectation values. The observed Z basis statistics matched the classical histogram while the corresponding expectation value was near 0, showing that there is an equal probability of obtaining $+1$ (for $|0\rangle$) or -1 (for $|1\rangle$). In contrast, the X basis expectation value was close to 1, showing that the same prepared state was an eigenstate of X.

The point of the lab was to show that $50-50$ statistics in the Z basis come from coherent superposition rather than from an incoherent classical mixture. This difference was shown by varying the relative phase. Two consecutive Hadamard gates returned the state to the computational ground state through constructive and destructive interference of amplitudes with a relative phase of π. Inserting a Z phase rotation of π between the two Hadamards inverted the final outcome. The interference was changed deterministically by the phase rotation, highlighting that the observed measurement statistics are set by both the prepared state and the measurement basis.

A geometric interpretation on the Bloch sphere was then used to visualize these observations. Single qubit gates were understood as rotations of a unit vector. The $\sqrt{X}$ gate rotated the vector by $\pi/2$ radians about the x axis and produced expectation values that were near 0 in X and Z and near -1 in Y, which is consistent with the vector pointing in the $-y$ direction. The Hadamard gate was shown to effectively swap x and z axes through identities such as $HZH = X$, and $HXH = Z$. These relations explained why a state that appears balanced in one basis can be an eigenstate in another basis. In this view the quantum coin is not a coin that is flipping in time, it is a vector that evolves deterministically under gates and only becomes random when it is projected onto a chosen axis during measurement.

The overall interpretation was that superposition is a coherent quantum state with basis dependent statistics, not a classical probabilistic mixture. The Bloch sphere was used to visualize single qubit states and to identify, in the chosen basis, which measurements are deterministic and which are balanced. The classical coin analogy was useful to set intuition but it was found to be incomplete because it lacks phase and cannot produce interference. A more complete image was a coin balanced on its edge with an orientation in three dimensional space, with the Bloch vector pointing in the direction of the coin's 'head', where gates rotate the orientation and measurement projects it. This work therefore demonstrated how phase, basis, and rotation identities together provide a predictive framework for preparing and reading single qubit states.

Stern-Gerlach Measurement

The purpose of this lab was to connect the Stern-Gerlach experiment with single qubit behavior on quantum hardware. The question was whether the discreteness seen in silver atom beams can be reproduced and interpreted using single qubit projective measurements. To begin, a qubit was initialized in an arbitrary state, and measurements were performed. Single shots returned only one of two outcomes, similar to the two beam spots of the original experiment, and repeated shots produced frequencies set by the squared amplitudes of the prepared state. This confirmed that for this arbitrary state, quantization is found, rather than a hidden continuous spread.

A distribution of randomly oriented spins was simulated to mirror an oven source. One measurement was taken per state and results were averaged. The distribution was seen to be roughly even, which is expected for uniform orientations, yet every individual measurement remained discrete.

The idea of the state collapsing into an eigenstate of the measurement basis was studied next. Two measurements were made in succession. After preparation in an equal superposition (specifically the state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$), the first readout fixed the state in an eigenstate of the Z basis, and the second readout, with measurement in the same axis, returned the same value (up to small noise effects). This behavior made clear that a measurement is not the same as applying a unitary operator. Acting with the Pauli Z changes a relative phase, retaining coherence, while a Z basis measurement is a projective, probabilistic collapse that is not reversible.

The role of measurement basis was then explored. By default, the device measures along the Z axis, so measurements along X were implemented by a basis change with a Hadamard before the readout. States prepared as $|0\rangle$, or $|1\rangle$ were found to give equal probabilities for the X basis outcomes, which was expected since $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$, and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$. Two consecutive measurements in the X basis reproduced the same collapse and repeatability that had been observed in Z. For this section, a statevector sampler was used, which gives ideal, noiseless probabilities from the statevector. A Bloch sphere description accounted for the results, rotations set the state orientation and projective measurement gave one of the eigenvalues along the selected axis. From the observed statistics, the X eigenstates were reconstructed in the Z basis and the matrix form of S_x was obtained by spectral decomposition.

This notebook showed that single qubit projective readout can reproduce the results of the Stern Gerlach experiment. The basis dependence of the measurement statistics was observed directly and explained on the Bloch sphere.

Exploring Uncertainty

The final lab demonstrated that quantum uncertainty is a fundamental property of quantum systems, and not a consequence of measurement limitations. This was shown by measuring outcome distributions and expectation values for non-commuting observables. Qubits were prepared with simple rotations, then read out after changes of basis.

To start, the role of measurement order was examined. When the first measurement was in Z, the initialized state was projected to a Z eigenstate (always '0' since qubits are by default initialized in $|0\rangle$), and a subsequent measurement in X gave balanced outcomes. When the order was reversed, the first measurement in X produced balanced outcomes and the second measurement in Z (by applying another Hadamard gate) again produced balanced outcomes. The four joint outcomes matched the expected distribution. This showed that a projective measurement collapses the state onto an eigenstate of the measured observable and removes coherence in that basis.

Uncertainties were then found using expectation values. For the state that points along the $+y$ direction, the expectation value of Y was near 1, while the expectation values of X and Z were near 0 $\Rightarrow \Delta X = \Delta Z = 1$, and $\Delta Y = 0$. These results matched the prediction for a state that is an eigenstate of Y, and the uncertainty relation $\Delta X \Delta Z = 1 \geq |\langle Y \rangle| = 1$ was satisfied, and at the lower bound.

Rotations R_y of the ground state through the xz plane confirmed the inequality $\Delta X \Delta Z \geq |\langle Y \rangle|$ at every angle. Adding a fixed R_z phase moved the state off that plane, produced nonzero Y expectations, and again the bound was satisfied and saturated at the expected angles (e.g. when $\theta = \pi/2$, where the state aligns with y). Using the same rotations, $\Delta X \Delta Y \geq |\langle Z \rangle|$ was tested, and confirmed to be satisfied. Trying to test the uncertainty relations further, the large $\langle Y \rangle$ limit was considered, a set of rotations were made within the yz plane, and $\Delta X \Delta Z \geq |\langle Y \rangle|$ was respected at each point (actually saturated), even at $\theta = \phi = \pi/2$, where $|\langle Y \rangle| = 1$ (an eigenstate of Y). Rotations over the full Bloch sphere were calculated, and a surface plot of the margin $\Delta X \Delta Z - |\langle Y \rangle|$ versus θ and ϕ was produced. This quantity should be non-negative everywhere if the uncertainty relation is satisfied, and equals 0 exactly where $\Delta X \Delta Z = |\langle Y \rangle|$. As expected, the margin was non-negative everywhere and 0 in the yz or xy plane.

Two further concepts were studied. First, measurements along intermediate directions were implemented by scanning linear combinations of Pauli operators. Expectation values changed smoothly with the rotation angle, showing that uncertainty is basis dependent. Second, simple tomography reconstructed the Bloch vector of a rotated state. The reconstructed components agreed with analytic values, validating the tomography.

This notebook showed that quantum uncertainty follows from Bloch sphere geometry and from the non-commutation of observables. Projective measurement collapses the state into an eigenstate of the measurement basis and simultaneously erases information about orthogonal components. The generalized uncertainty relations were satisfied for each state studied, and were found to be saturated at the predicted states.